

Notes: Giannetti and Wang (JF, 2016)

Corporate Scandals and Household Stock Market Participation

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1 Big picture

- **Question.** Does the revelation of corporate financial fraud reduce households' willingness to participate in the stock market? More specifically, when fraud is revealed in a household's state, do local households cut equity participation and equity holdings?
- **Core idea.** Corporate fraud may do more than destroy value in the fraudulent firm. It may also reduce *trust* in financial markets and thereby shrink household demand for equity more broadly.
- **Why the local angle matters.** The authors assume households are more exposed to fraud committed by firms headquartered in their own state because local firms receive more local media coverage and are more salient through social and business networks. This gives cross-state and within-state variation in exposure to fraud.
- **Main data.** The paper combines:
 - a database of 704 federal securities-fraud enforcement actions involving 695 U.S. firms from 1980 to 2009;
 - the PSID household survey for stock market participation, wealth, and demographics;
 - brokerage-account data to see which stocks households actually buy and sell;
 - Gallup confidence surveys to test whether fraud reduces trust in big business and banks.
- **Main baseline finding.** Fraud revelation in a state is followed by a decline in local household stock market participation on both the extensive and the intensive margin. A one-standard-deviation increase in four-year fraud intensity lowers the probability of stock market participation by about 1.23 percentage points, increases exit by about 2.6 percentage points, decreases entry by about 0.8 percentage points, and lowers the equity share of household wealth.
- **Main identification finding.** Using Arthur Andersen's collapse as an arguably exogenous shock to fraud revelation, the paper finds even larger effects. A one-standard-deviation increase in fraud revelation induced by the Andersen shock raises the probability of exiting the stock market by roughly seven percentage points and sharply reduces new equity purchases.
- **Main mechanism finding.** The evidence is most consistent with a *loss of trust* rather than with a direct wealth effect or rational information spillovers. Fraud lowers trust in big business and banks, and the reaction is stronger for groups that the authors interpret as having higher betrayal costs.
- **Not just holders of fraudulent stocks.** Using brokerage data, the paper shows that households cut holdings even when they did *not* own the fraudulent stocks, and even in nonfraudulent firms outside the fraudulent firms' industries and states.
- **Not just local bad news about fundamentals.** Households cut *out-of-state* stocks more than in-state stocks after local fraud revelation, implying an increase in local bias and market segmentation rather than simply updated beliefs about local firms' cash flows.
- **Cross-sectional mechanism evidence.** The response to fraud is stronger for high-status, middle-aged, and college-educated households, but not stronger for low-risk-tolerance or high-wealth households. This supports a betrayal-cost interpretation rather than a simple risk-aversion story.

- **Corporate-finance implication.** Nonfraudulent firms located in states hit by fraud lose retail ownership, repurchase more shares, and become more concentrated in the hands of large institutional investors. This is suggestive evidence that fraud raises the cost of capital for innocent firms.
- **Economic takeaway.** Corporate fraud generates a negative externality: it does not only hurt shareholders of the fraudulent firm, it also reduces trust and therefore depresses aggregate household demand for equity.

2 Data and measurement (key objects)

2.1 Fraud data

- **Source.** Federal Securities Regulation (FSR) database compiled by Karpoff, Lee, and Martin.
- **What counts as fraud.** The sample includes federal enforcement actions for intentional and material financial misrepresentation involving U.S. issuers and common stock under the Securities Act of 1933 or the Exchange Act of 1934.
- **Sample size.** 704 cases involving 695 U.S. firms from 1980 to 2009.
- **Timing.** A fraud’s revelation year is the earliest announcement date of any key litigation event associated with the case.
- **Location.** Firms are assigned to states using headquarters data from Compustat and Compact Disclosure, with missing information hand-collected.
- **Geographic variation.** Fraud is widespread but concentrated: California, New York, Texas, and Florida have many cases, yet only five states have no federal securities-fraud litigation over the sample period.
- **Time variation.** Fraud cases spike during the accounting-scandal years around 2000–2002, which provides useful time-series variation.

2.2 Main exposure variables

- **Baseline local fraud exposure.**

$$Fraud\ in\ State_{jt} = \frac{\text{number of frauds revealed in state } j \text{ over the relevant window}}{\text{number of listed firms in state } j}.$$

- **Why divide by listed firms?** Large states have more firms and therefore mechanically more fraud cases. Scaling by the number of public firms makes fraud intensity comparable across states.
- **Relevant window.**
 - In PSID specifications, the baseline measure cumulates fraud over the previous four years because household wealth data are observed every five years early in the sample and every two years later.

- In brokerage and Gallup specifications, the paper uses yearly fraud revelation because those data are annual.

- **Alternative exposure measures.**

- **Fraud in State (dummy):** at least one fraud in the state over the previous four years.
- **Fraud in State-Local Firms:** larger weights for fraud by firms operating in no more than two states.
- **Fraud in State-Local Firms & High Media Coverage:** larger weights for geographically concentrated firms whose fraud received more news coverage.
- **Yearly Fraud in Other States:** fraud outside the household's home state, used to benchmark local versus nonlocal responses.
- **Experienced Fraud:** a household's average lifetime exposure to fraud since age 18.

2.3 PSID household survey data

- **Source.** Panel Study of Income Dynamics (PSID), which records households' state of residence, financial wealth, equity holdings, and demographics.
- **Survey years used.** 1984, 1989, 1994, 1999, 2001, 2003, 2005, 2007, and 2009.
- **Main extensive-margin outcome.**
 - **Equity Participation:** indicator equal to one if the household holds stocks, mutual funds, or investment trusts.
 - **Equity Participation (IRA):** same idea but consistently including IRA and pension holdings.
- **Main intensive-margin outcomes.**
 - **Equity-Wealth Ratio:** net equity divided by total household wealth.
 - **Net Equity Purchases:** net dollar value of new equity investment during the year.
- **Transition outcomes.**
 - **Entry:** household enters the stock market between survey rounds.
 - **Exit:** household leaves the stock market between survey rounds.
- **Other portfolio outcome.** A dummy for holding bonds and other fixed-income securities is used to see whether fraud triggers substitution away from equity.
- **Key controls.** Age, marital status, family size, income, wealth excluding equity, college education, state GDP growth, and state stock returns.

Key PSID summary statistics.

- Equity participation: 31.3%.
- Equity participation including IRAs: 43.2%.
- Equity-wealth ratio: 6.6%.

- Entry rate: 13.3%.
- Exit rate: 28.3%.
- Average age of household head: 50.3 years.
- College-educated household heads: 27.7%.
- Households ever moving across states: 16.9%.

Interpretation. The PSID gives the authors repeated household-level participation outcomes together with the state location needed to match local fraud exposure.

2.4 Brokerage data

- **Source.** Discount-brokerage data originally used by Barber and Odean.
- **Why these data matter.** PSID does not reveal which stocks a household owns. Brokerage data let the authors test whether households sell only fraudulent firms or sell equity more broadly.
- **Sample.** More than 30,000 households from 1991 to 1996, with holdings in common stocks and mutual funds.
- **Main outcome.** Yearly change in equity holdings for different portfolios of stocks: all stocks, nonfraudulent stocks, stocks outside the fraudulent industry, stocks outside the fraudulent state, in-state stocks, and out-of-state stocks.
- **Brokerage sample facts.**
 - average yearly change in all equity holdings: 9.0%;
 - average yearly change in out-of-state holdings: 12.7%;
 - average yearly change in in-state holdings: 1.8%;
 - average yearly portfolio return: 16.0%.

2.5 Gallup confidence surveys

- **Source.** Gallup “Confidence in Institutions” surveys from 1981 to 2009.
- **Unit of analysis.** State-by-survey-year changes in the fraction of respondents reporting high confidence in an institution.
- **Main trust outcomes.**
 - Trust in Big Business;
 - Trust in Banks;
 - an average trust measure for other institutions, used as a control for broad shifts in trust unrelated to financial markets.
- **Why this matters.** The Gallup data provide direct evidence on the mechanism: whether fraud revelation reduces confidence in financial institutions.

2.6 State-level controls and the Arthur Andersen shock

- State GDP growth from the Bureau of Economic Analysis.
- State stock return computed from buy-and-hold returns of publicly traded firms headquartered in the state.
- **Arthur Andersen shock (AA Shock)**. Fraction of public firms in a state audited by Arthur Andersen in 2001–2002 and therefore forced to switch auditors after Andersen’s collapse.

Interpretation. These variables help separate the effect of fraud revelation from local business-cycle conditions and provide an instrument for arguably exogenous fraud detection.

3 Empirical design and specifications (all equations + interpretation)

3.1 Brokerage-data outcome: changes in equity holdings

For household i in year t , and for a portfolio of stocks J , the change in holdings is

$$\Delta \text{holding}_{i,t+1}^J = \frac{\sum_{k \in J} p_{kt} (\text{holding}_{ik,t+1} - \text{holding}_{ik,t})}{\sum_{k \in J} p_{kt} \text{holding}_{ik,t}}. \quad (1)$$

Interpretation.

- p_{kt} is stock k ’s price at the end of year t .
- The numerator values the change in the household’s position in each stock using the beginning-of-period price.
- The denominator scales the change by the initial value of the household’s portfolio in that group of stocks.
- This turns trading into a clean percentage change in holdings, comparable across households and portfolios.

3.2 Baseline household participation regression

The paper’s main empirical specification is

$$\text{Participation}_{ijt} = \beta \text{Fraud in State}_{jt} + \gamma X_{ijt} + s_j + \zeta_t + \varepsilon_{ijt}, \quad (2)$$

where household i lives in state j at time t .

Dependent variables.

- Extensive margin: Equity Participation, Entry, Exit.
- Intensive margin: Equity-Wealth Ratio, Net Equity Purchases.

Controls.

- household demographics and resources: age, marital status, family size, income, and wealth;
- education when applicable;
- state business conditions: GDP growth and state stock returns;
- in some specifications, lagged participation or household fixed effects.

Interpretation of β .

- If $\beta < 0$, more local fraud revelation is associated with lower local stock-market participation.
- Because year fixed effects absorb nationwide shocks and national fraud waves, the estimated β is the *differential local effect*. The authors therefore interpret the estimate as a lower bound on the total effect of fraud on aggregate equity demand.

3.3 Why OLS rather than nonlinear panel models?

- The outcomes include both continuous and indicator variables.
- The specifications contain many fixed effects and lagged dependent variables.
- The authors therefore estimate all equations by ordinary least squares and cluster standard errors at the household level.

Interpretation. This is mainly a practical fixed-effects linear probability strategy; the economic content comes from sign, magnitude, and robustness, not from a structural nonlinear probability model.

3.4 Dynamic and alternative-exposure specifications

The paper also estimates:

- models with lagged stock market participation to absorb persistence in household behavior;
- first-difference models where the dependent variable is **Exit**, **Entry**, or $\Delta(\text{Equity-Wealth Ratio})$;
- yearly-fraud models in the post-1999 sample to study short-run dynamics;
- variants that replace the baseline fraud proxy with the dummy, local-firm, and local-firm-plus-media measures.

Interpretation. These checks ask whether the result depends on how fraud exposure is measured or on whether the regression focuses on levels or changes.

3.5 Instrumental-variables strategy: Arthur Andersen

The first-stage idea is

$$\text{Fraud in State}_{jt} = \pi \text{AA Shock}_{jt} + \delta Z_{jt} + s_j + \zeta_t + u_{jt}. \quad (3)$$

Construction of the instrument.

- Arthur Andersen collapsed after Enron.
- Andersen’s clients had to switch auditors.
- New auditors had incentives to “clean house,” making fraud revelation more likely among ex-Andersen clients.
- States with a larger share of Andersen clients should therefore see a larger temporary increase in fraud revelation.

Why this helps.

- It isolates variation in fraud revelation generated by auditor turnover rather than by underlying local economic conditions.
- The paper also constructs pseudo-instruments using the client shares of the other Big 5 auditors. These pseudo-instruments do not predict fraud revelation, which supports the exclusion restriction.

3.6 Within-state lifetime exposure to fraud

To exploit variation across households living in the same state at the same time, the paper constructs lifetime fraud experience as an average of past fraud intensity since adulthood:

$$\text{Experienced Fraud}_{it} = \frac{1}{M_{it}} \sum_{k=1}^{M_{it}} \text{Fraud}_{s(i,t-k),t-k}, \quad M_{it} = \min\{\text{age}_{it} - 18, t - 1980\}. \quad (4)$$

Interpretation.

- Two households currently in the same state can have different fraud experience because they differ in age or because they have moved across states.
- This lets the authors include **state** $\times$ **year** fixed effects, which absorb all current state-level shocks.
- If experienced fraud still predicts stockholding, the effect is hard to explain using only contemporaneous local macro conditions.

3.7 Trust regressions

At the state level, the trust tests relate changes in confidence to yearly fraud revelation:

$$\Delta \text{Trust}_{jt} = \theta \text{Yearly Fraud in State}_{j,t-1} + \rho \Delta Z_{jt} + \zeta_t + \eta_{jt}. \quad (5)$$

Interpretation.

- If $\theta < 0$, fraud revelation reduces confidence in big business or banks.
- This is the paper’s direct evidence that trust is part of the causal chain from fraud to lower equity participation.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the design is plausible

- **Local salience.** Fraud by local firms is more likely to be seen, discussed, and remembered by local households.
- **State variation.** Fraud intensity varies substantially across states and over time.
- **Household panel structure.** The PSID allows the authors to compare the same household over time, or households in the same state-year with different prior exposure histories.

4.2 Threat 1: local downturns drive both fraud revelation and stock sales

- Fraud may be revealed near recessions or state-level downturns.
- The paper therefore controls for state GDP growth and state stock returns.
- It also uses household fixed effects and lagged dependent variables.
- Stronger still, the lifetime-experience regressions include **state** $\times$ **year** fixed effects, eliminating all contemporaneous state shocks.

Why this matters. If the result survives these controls, it is difficult to explain it purely with local business-cycle conditions.

4.3 Threat 2: direct wealth effect from holding fraudulent stocks

- If households lose money on the fraudulent stock, they may mechanically hold less equity afterward.
- Brokerage tests show that households *without* holdings in fraudulent firms also cut equity.
- The effect persists when the dependent variable excludes fraudulent stocks entirely.

Conclusion. The main effect is not just a mechanical balance-sheet loss.

4.4 Threat 3: rational information spillovers about related firms

- Fraud revelation could rationally change beliefs about other firms in the same industry or same state.
- The paper therefore excludes stocks in the same two-digit SIC industries as the fraudulent firms and still finds selling.
- It also excludes firms headquartered in states where fraud was revealed and still finds selling.
- Households reduce out-of-state stocks more than in-state stocks, which points toward trust/familiarity effects rather than local information spillovers.

4.5 Threat 4: omitted state characteristics correlated with Arthur Andersen clients

- States with many Andersen clients might differ in unobserved ways.
- The paper constructs placebo instruments using client shares of Deloitte, Ernst & Young, KPMG, and PricewaterhouseCoopers.
- These placebo instruments do not predict fraud revelation after 2001, whereas Andersen exposure does.

Conclusion. The identifying power comes from forced auditor switching after Andersen’s collapse, not from generic features of states with Big 5 clients.

4.6 Threat 5: fraud simply raises risk aversion

- If fraud mainly makes households more risk averse, the effect should be strongest for households that were already less risk tolerant.
- The interaction tests do not support this. The effect is not significantly stronger for low-risk-tolerance households or for low-wealth households.
- Instead, the effect is stronger for groups interpreted as having higher betrayal costs.

Conclusion. The mechanism looks more like trust/betrayal than pure changes in standard risk preferences.

4.7 What remains imperfect

- Exposure is measured at the state level rather than at a finer geographic level.
- Gallup does not let the authors follow the same individual over time, so trust evidence is state-level rather than household-level.
- The paper documents changes in investor base consistent with higher cost of capital, but does not directly estimate firms’ cost-of-capital effects.
- Because year fixed effects absorb nationwide scandals, the estimated coefficients are lower bounds on the total effect of fraud on stock demand.

5 Tables and results

5.1 Table I: corporate financial fraud by state and year

Read: This table shows where and when fraud cases occur.

- Fraud is concentrated in states with many public firms: California has 127 cases, New York 84, Texas 71, and Florida 67.
- But fraud is not confined to a handful of states; only five states have no cases over the sample.

Table I
Corporate Financial Fraud by State and Year
This table reports the number of corporate security frauds revealed between 1980 and 2009 by state and by year. The year is the year in which a fraud case is first revealed. Detected corporate securities frauds come from the Federal Securities Regulation (FSR) database, compiled by Karpoff, Lee, and Martin (2014).

Panel A: Fraud Cases by State							
State	# of Frauds	State	# of Frauds	State	# of Frauds	State	# of Frauds
AL	5	IA	1	MN	12	NY	84
AR	1	ID	1	MO	8	OH	25
AZ	10	IL	19	MS	2	OK	5
CA	127	IN	8	MT	1	OR	3
CO	16	KS	10	NC	8	PA	24
CT	18	KY	1	NE	3	RI	1
DC	2	LA	2	NH	2	SC	5
DE	2	MA	29	NJ	29	SD	3
FL	67	MD	8	NM	2	TN	6
GA	22	MI	12	NV	12	TX	71

Panel B: Fraud Cases by Year					
Year	# of Frauds	Year	# of Frauds	Year	# of Frauds
1980	4	1988	14	1996	31
1981	6	1989	12	1997	23
1982	11	1990	16	1998	29
1983	12	1991	21	1999	29
1984	17	1992	24	2000	51
1985	11	1993	27	2001	38
1986	12	1994	40	2002	59
1987	23	1995	26	2003	25

- Time-series clustering is also clear: fraud revelations surge in the late 1990s and early 2000s, with especially high counts in 2000, 2002, and 2006.

Why it matters. The table establishes the raw geographic and time variation that the paper exploits. It also shows why scaling by listed firms is necessary: some states mechanically host far more firms and hence more frauds.

5.2 Table II: summary statistics

Table II
Summary Statistics
Panel A reports summary statistics for the household- and state-level variables of the PSID sample. The survey years are 1984, 1989, 1994, 1999, 2001, 2003, 2005, 2007, and 2009. Summary statistics for the PSID sample and all estimations involving this sample are reported using sample population weights to be representative of the underlying population. Panel B reports summary statistics for the household- and state-level variables of the brokerage sample. The sample period is 1991 to 1996. Panel C reports the state-level data for the Gallup Survey sample. The survey years include 1981, 1983, 1984, 1985, 1986, 1990, 1991, and 1993 to 2010.

Panel A: PSID Sample			
Variable	N	Mean	SD
<i>Household-Level Data (by household and survey year)</i>			
Equity Participation	66,429	0.313	0.464
Equity Participation (IRA)	66,429	0.432	0.495
Equity-Wealth Ratio	66,370	0.066	0.161
Entry	40,528	0.133	0.340
Exit	12,062	0.283	0.451
Net Equity Purchases	65,354	11,291	105,569
Bonds & Fixed Income Securities	17,031	0.167	0.373
Age (household head)	66,615	50.25	16.60
College Education	64,540	0.277	0.448
Experienced Fraud	65,253	0.008	0.006
Experienced State GDP Growth	65,253	0.093	0.035
Experienced State Stock Return	65,253	0.066	0.015
Family Size	66,429	2.468	1.394
Family Income (in thousands)	65,929	62.62	93.67
High Status	66,429	0.278	0.448
Low Risk Tolerance	26,106	0.645	0.478
Married	66,429	0.585	0.493
Middle Aged (household head)	66,429	0.614	0.487
Moved	66,359	0.046	0.211
Wealth (excl. equity, in thousands)	66,408	215.1	1,368.8
<i>Household-Level Data (by household)</i>			
Mover	13,839	0.169	0.375
<i>State-Level Data (by state and survey year)</i>			
AA Shock	204	0.105	0.128
Fraud in State	459	0.011	0.022
Fraud in State (dummy)	459	0.546	0.498
Fraud in State-Local Firms	459	0.013	0.024
Fraud in State-Local Firms & High Media Coverage	459	0.019	0.039
Yearly Fraud in State	459	0.002	0.006
Yearly Fraud in Other States	459	0.012	0.004
State GDP Growth	459	0.058	0.024
State Stock Return	459	0.390	0.628

(Continued)

Table II—Continued			
Panel B: Brokerage Sample			
Variable	N	Mean	SD
<i>Household-Level Data (by household and year)</i>			
Change in Equity Holdings (all stocks)	127,176	0.090	0.338
Change in Holdings (excl. fraudulent stocks)	126,878	0.091	0.338
Change in Holdings (only stocks of firms in industries with no fraud revelation)	125,528	0.088	0.336
Change in Holdings (only stocks of firms in states with no fraud revelation)	80,611	0.090	0.334
Change in Holdings (only out-of-state stocks)	63,727	0.127	0.379
Change in Holdings (only in-state stocks)	63,727	0.018	0.125
Household Portfolio Return	127,176	0.160	0.420
Age	127,176	42	23
Married	127,176	0.508	0.500
Family Size	127,176	2.264	1.119
<i>State-Level Data (by state and year)</i>			
Yearly Fraud in State	255	0.002	0.005
Yearly Fraud in State (dummy)	255	0.290	0.455
Yearly Fraud in Other States	255	0.112	0.043
State GDP Growth	255	0.057	0.030
Panel C: Gallup Survey Sample			
<i>State-Level Data (by state and survey year)</i>			
Δ(Trust in Big Business)	1,126	0.000	0.195
Δ(Trust in Banks)	1,126	-0.012	0.243
Δ(Trust in Other Institutions)	1,126	-0.004	0.193
Δ(Log(Age))	1,126	0.011	0.181
Δ(College Education)	1,126	0.009	0.217
Δ(High Income)	956	0.010	0.218
Δ(Male)	1,126	-0.000	0.205
Δ(Protestant)	577	-0.005	0.221
Δ(Jewish)	577	0.002	0.052
Δ(Republican)	1,126	0.001	0.215
Δ(White)	1,126	-0.001	0.195
Δ(Black)	1,126	-0.000	0.089
Δ(Hispanic)	795	-0.000	0.089
Yearly Fraud in State	1,126	0.003	0.009
State GDP Growth	1,126	0.054	0.035
State Stock Return	1,126	0.085	0.232

Read: This table describes the PSID sample, the brokerage sample, and the Gallup trust sample.

- In the PSID, only 31.3% of households hold equity outside IRAs, while 43.2% hold equity once IRAs are included.
- The average equity share of total wealth is only 6.6%.
- Entry into the stock market is infrequent (13.3%), and exit among participants is common (28.3%).
- In the brokerage sample, yearly changes in stockholdings are much larger for out-of-state than in-state stocks, which becomes important later when the paper studies rebalancing.

Why it matters. The summary statistics show that nonparticipation is economically important and that there is plenty of room for fraud revelation to affect participation decisions.

5.3 Table III: fraud revelation and household stock market participation

Read: This is the paper’s main reduced-form evidence.

Table III						
Fraud Revelation and Household Stock Market Participation						
This table relies on the PSID sample. In Panel A, we relate the proxy for fraud revelation intensity, cumulated over a four-year period, to household stock market participation at the end of the period and changes in stock market participation during that period. In column (4), we exclude observations from the 1984 and 1989 surveys, for which we are unable to separate any equity held in IRAs from the remaining equity holdings. In column (5), lagged equity participation includes equity in IRAs. In columns (6) and (7), all household characteristics are expressed in first differences, as the dependent variable is a change. Parameter estimates are obtained by ordinary least squares. Panel B explores the extent to which alternative proxies for a household's exposure to local fraud are related to household stock market participation. <i>Fraud in State-Local Firms</i> gives larger weights to frauds committed by local firms (defined as firms that operate in no more than two states), and <i>Fraud in State-Local Firms & High News Coverage</i> gives larger weights to frauds that received higher news coverage, committed by local firms (defined as above). The numbers in square brackets in columns (2) and (3) are the impact of a one-standard-deviation change in the relevant <i>Fraud in State</i>. In columns (4) and (5), we exclude survey rounds carried out before 1999 and relate equity participation to fraud revelation intensity during the past year; in column (5) we also include one lag. In Panel C, we relate the proxy for fraud revelation intensity, cumulated over a four-year period, to household stock market participation on the intensive margin at the end of the period and changes in the continuous proxies for stock market participation during that period. The dependent variable is indicated at the top of each column. In column (4), we exclude observations from the 1984 and 1989 surveys, for which we are unable to separate any equity held in IRAs from the remaining equity holdings. In column (5), the lagged equity-wealth ratio includes equity in IRAs. In column (6), all household characteristics are expressed in first differences, as the dependent variable is a change. All variables are defined in the Appendix. Parameter estimates are obtained by OLS. All regressions include a constant term and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the household level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.						
Panel A: Stock Market Participation on the Extensive Margin						
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Equity Participation	Equity Participation	Equity Participation	Equity Participation (excl. 1984 & 1989)	Equity Participation (with IRA)	Exit	Entry
	(0.007)	(0.009)	(0.005)	(0.005)	(0.005)	(0.008)
Log(Family Income)	0.048***	0.014***	0.034***	0.020***	-0.018***	0.003*
Log(Wealth)	0.003	(0.002)	(0.002)	(0.002)	(0.006)	(0.001)
State GDP Growth	0.029***	0.014***	0.019***	0.019***	-0.030***	0.005***
State Stock Return	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)
College Education	0.286	0.399**	0.177	-0.139	-0.138	-0.259
Equity Participation (lagged)	(0.169)	(0.191)	(0.189)	(0.206)	(0.181)	(0.240)
Household F.E.	0.002	0.001	0.008	0.007	0.005	-0.004
Year F.E., State F.E.	(0.008)	(0.009)	(0.007)	(0.007)	(0.006)	(0.017)
Observations	0.207**	(0.011)	0.454***	0.489***	0.291***	
R ²			(0.008)	(0.008)	(0.007)	
	Yes	Yes	Yes	Yes	Yes	Yes
	64,191	66,084	52,257	47,121	52,257	40,157
	0.239	0.581	0.398	0.403	0.719	0.062
Panel B: Alternative Measure of Local Exposure to Fraud						
(1)	(2)	(3)	(4)	(5)		
Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation (after 1999)	Equity Participation (after 1999)	Equity Participation (after 1999)
Fraud in State (dummy)			-0.031***			

(Continued)

(Continued)

Panel A: Extensive margin.

- The baseline coefficient on **Fraud in State** is about -0.53 to -0.60 across specifications for equity participation.
- The preferred specification with lagged participation implies that a one-standard-deviation increase in four-year fraud intensity (2.2 percentage points) lowers equity participation by about 1.23 percentage points.
- This is roughly a 4% decline relative to the 31% participation rate in the sample.
- Fraud also increases the probability of **Exit** by about 2.6 percentage points and lowers **Entry** by about 0.8 percentage points.

Interpretation. Fraud makes households already in the market more likely to leave, and households outside the market less likely to enter.

Panel B: Alternative exposure measures.

Table III—Continued

Panel B: Alternative Measure of Local Exposure to Fraud				
(1)	(2)	(3)	(4)	(5)
	Equity Participation	Equity Participation	Equity Participation	Equity Participation
	(0.007)		(after 1999)	(after 1999)
Fraud in State-Local Firms		−0.558*** (0.191) [−0.015]		
Fraud in State-Local Firms & High-Media Coverage			−0.311*** (0.088) [−0.033]	
Yearly Fraud in State			−0.545** (0.270)	−0.886*** (0.219)
Lagged Yearly Fraud in State				0.196 (0.200)
Log(Age)	−0.030*** (0.007)	−0.014** (0.008)	−0.014** (0.006)	0.005 (0.007)
Married	0.034*** (0.006)	0.028*** (0.007)	0.028*** (0.005)	0.025*** (0.007)
Log(Family Size)	−0.034*** (0.005)	−0.037*** (0.005)	−0.037*** (0.004)	−0.037*** (0.005)
Log(Family Income)	0.028*** (0.002)	0.034*** (0.002)	0.028*** (0.002)	0.028*** (0.002)
Log(Wealth)	0.020*** (0.001)	0.018*** (0.001)	0.018*** (0.000)	0.019*** (0.001)
State GDP Growth	1.052*** (0.187)	0.174 (0.180)	0.148 (0.145)	0.057 (0.242)
State Stock Return	0.001 (0.005)	0.008 (0.007)	0.008 (0.006)	0.007 (0.013)

(Continued)

Table III—Continued

Panel E: Alternative Measure of Local Exposure to Fraud				
(1)	(2)	(3)	(4)	(5)
	Equity Participation	Equity Participation	Equity Participation	Equity Participation
				(after 1999)
Lagged Equity Participation	0.488*** (0.046)	0.484*** (0.048)	0.484*** (0.044)	0.527*** (0.009)
Year F.E., State F.E.	Yes	Yes	Yes	Yes
Observations	52,257	52,257	52,257	36,441
R ²	0.393	0.398	0.398	0.412

Panel C: Participation on the Intensive Margin						
(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log (Net Equity Purchases)	Equity-Wealth Ratio	Equity-Wealth Ratio	Equity-Wealth Ratio (excl. 1984, 1989)	Equity-Wealth Ratio (with IRA)	Bonds & Fixed Income Securities
Fraud in State	−2.527* (1.531)	−0.191** (0.087)	−0.196*** (0.072)	−0.190** (0.075)	−0.210*** (0.071)	−0.269*** (0.109)
Fraud in State *						−0.202 (0.541)
Lagged Equity Participation						0.997*** (0.567)
Log(Age)	1.349 (0.962)	0.006 (0.026)	0.012*** (0.003)	0.013*** (0.003)	0.010*** (0.003)	0.032* (0.018)
Married	−0.143 (0.125)	0.001 (0.005)	0.003 (0.003)	0.003 (0.003)	−0.002 (0.003)	−0.003 (0.005)
Log(Family Size)	0.161* (0.095)	0.000 (0.003)	0.012*** (0.002)	−0.011*** (0.002)	−0.006*** (0.002)	0.002 (0.004)
Log(Family Income)	0.046* (0.024)	0.002*** (0.001)	0.008*** (0.001)	0.008*** (0.001)	0.006*** (0.001)	0.001 (0.001)
Log(Wealth)	0.033*** (0.006)	0.004*** (0.000)	0.006*** (0.000)	0.006*** (0.000)	0.004*** (0.000)	0.012*** (0.001)

(Continued)

Table III—Continued

Panel C: Participation on the Intensive Margin						
(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Log (Net Equity Purchases)	Equity-Wealth Ratio	Equity-Wealth Ratio	Equity-Wealth Ratio (excl. 1984, 1989)	Equity-Wealth Ratio (with IRA)	Bonds & Fixed Income Securities
State GDP Growth	0.057 (2.630)	−0.128* (0.069)	−0.206*** (0.069)	−0.176** (0.081)	−0.202*** (0.082)	0.079 (0.401)
State Stock Return	0.052 (0.085)	0.005 (0.003)	0.005 (0.003)	0.005* (0.003)	0.002 (0.004)	0.005 (0.018)
Equity-Wealth Ratio (lagged)			0.386*** (0.013)	0.386*** (0.014)		
Bonds (lagged)						−0.050*** (0.017)
Household F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E., State F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	65,012	66,047	52,205	47,092	52,249	13,827
R ²	0.325	0.471	0.216	0.221	0.282	0.435

- A simple dummy for any fraud in the state implies an over 3-percentage-point decline in participation.
- Weighting local firms more heavily gives a one-standard-deviation effect of about 1.5 percentage points.
- Weighting local firms with high media coverage yields an even larger effect: about 3.3 percentage points.
- Using yearly fraud after 1999 also gives a negative effect, and the lagged yearly fraud variable is not significant, indicating little evidence of a quick reversal.

Interpretation. The result is not an artifact of one specific fraud measure. It becomes stronger when the exposure proxy better captures truly local and salient fraud.

Panel C: Intensive margin.

- Fraud reduces **Log(Net Equity Purchases)** and lowers the **Equity-Wealth Ratio** by roughly 0.19 to 0.21 in the main specifications.
- Changes in the equity-wealth ratio are also negative.
- Households previously holding stocks increase their holdings of bonds and fixed-income securities after fraud revelation.

Interpretation. Households do not merely stop entering the market; they also rebalance existing portfolios away from equity toward safer assets.

Why it matters. Table III is the central fact of the paper: local fraud revelation depresses local demand for equity on several margins.

5.4 Table IV: identification through the Arthur Andersen shock

Table IV
Identification through an Exogenous Shock to Fraud Revelation

In Panel A, the dependent variable is *Fraud in State*, the measure of fraud revelation intensity over the past four years in a state. The unit of observation is the state-year. The sample period is 2001 to 2007. Since *Fraud in State* is backward looking, we set the value of *AA Shock* equal to the fraction of public firms in a state that were AA clients and had to change auditors in 2001 and 2002 for the years between 2003 and 2005, and to zero for remaining years. The pseudo instrument (XX) is created in the same way except that we use the fraction of public firms that were clients of auditing firm XX in 2001 and 2002. *State GDP Growth* and *State Stock Return* are defined in the Appendix. Panels B and C rely on the PSID sample for the period 2001 to 2007. In Panel B, we present the first stage of the IV estimates for the effect of the instrument *AA Shock* on the endogenous variable *Fraud in State* in levels (column (1)) and first differences (column (2)). When we consider the equations in first differences, all control variables are expressed in first differences. In Panel C, we present two-stage least squares estimates for the effect of fraud revelation on alternative measures of stock market participation in levels and first differences. In particular, column (1) presents the first-stage estimates of the effect of the instrument on *Fraud in State* for the equations in levels. The second-stage estimates are presented in columns (3) to (5) and the dependent variable is indicated at the top of each column. Column (2) presents the first-stage estimates for the equations in first differences; the second-stage estimates are presented in columns (6) to (8). When we consider the equations in first differences, all control variables are expressed in first differences. Parameter estimates are obtained by OLS. All regressions include a constant term and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the year level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

Panel A: Validity of the Instrument					
	(1)	(2)	(3)	(4)	(5)
	Fraud in State				
AA Shock	0.048** (0.019)				
Pseudo Instrument (Deloitte & Touche)		-0.003 (0.003)			
Pseudo Instrument (Ernst & Young)			-0.007 (0.004)		
Pseudo Instrument (KPMG)				-0.025 (0.019)	
Pseudo Instrument (PWC)					0.000 (0.013)
State GDP Growth	0.024 (0.182)	0.035 (0.200)	0.036 (0.200)	0.034 (0.198)	0.036 (0.201)
State Stock Return	-0.003 (0.003)	-0.003 (0.003)	-0.003 (0.003)	-0.004 (0.003)	-0.003 (0.003)
Year F.E.	Yes	Yes	Yes	Yes	Yes
Observations	255	255	255	255	255
R ²	0.098	0.069	0.069	0.070	0.071

(Continued)

Table IV—Continued

Panel B: First-Stage IV Estimates						
	(1)	Fraud in State		(2)		
		First Stage				
<i>Instrumental Variable</i>						
AA Shock		0.026** (0.002)		0.025** (0.002)		
<i>Control Variables</i>						
Log(Age)		0.000 (0.000)		-0.015** (0.001)		
Married		0.000 (0.000)		0.000 (0.000)		
Log(Family Size)		-0.000* (0.000)		0.000 (0.000)		
Log(Family Income)		0.000** (0.000)		0.000** (0.000)		
Log(Wealth)		-0.000 (0.000)		0.000 (0.000)		
State GDP Growth		0.128** (0.010)		0.135** (0.011)		
State Stock Return		-0.003** (0.000)		-0.002** (0.000)		
College Education		0.000 (0.000)				
State F.E., Year F.E.		Yes		Yes		
Observations		38,841		33,603		
R ²		0.045		0.047		

Panel C: Second-Stage IV Estimates						
	(1)	(2)	(3)	(4)	(5)	(6)
	Equity Participation	Equity- Wealth Ratio	Log (Net Equity Purchases)	Exit	Entry	Δ(Equity- Wealth Ratio)
Fraud in State	-1.073* (0.624)	-0.600*** (0.232)	-14.533** (6.530)	4.905** (2.191)	2.302 (1.837)	-0.522 (0.508)
Log(Age)	0.052*** (0.007)	0.026*** (0.003)	0.082** (0.038)	1.846*** (0.265)	-0.070 (0.049)	-0.004 (0.032)
Married	0.056*** (0.006)	0.006*** (0.002)	0.146*** (0.035)	-0.073** (0.035)	0.007 (0.007)	-0.009 (0.006)
Log(Family Size)	-0.061*** (0.005)	-0.014*** (0.002)	-0.137*** (0.027)	0.025 (0.034)	0.004 (0.005)	0.003 (0.004)
Log(Family Income)	0.036*** (0.002)	0.007*** (0.001)	0.126*** (0.013)	-0.005 (0.009)	0.002** (0.001)	0.001 (0.001)
Log(Wealth)	0.022*** (0.001)	0.006*** (0.000)	0.060*** (0.003)	-0.026*** (0.003)	0.004*** (0.000)	0.003*** (0.000)
State GDP Growth	0.688*** (0.152)	0.108** (0.054)	2.418* (1.435)	-1.582** (0.461)	0.011 (0.243)	0.081 (0.102)
State Stock Return	-0.031*** (0.006)	-0.006** (0.002)	-0.042 (0.065)	0.041* (0.023)	0.045* (0.024)	0.005 (0.005)
College Education	0.153*** (0.007)	0.033*** (0.002)	0.359*** (0.035)			
State F.E., Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	38,841	38,841	38,841	6,401	22,075	28,389
R ²	0.214	0.109	0.034	0.019	0.012	0.005

Read: This table uses Andersen's collapse as an exogenous shock to fraud detection.

Panel A: Validity of the instrument.

- AA Shock significantly predicts later fraud revelation in the state.
- The coefficient is about 0.048, statistically significant at the 5% level.
- Pseudo-instruments based on the other Big 5 auditors do not predict fraud revelation.

Interpretation. The evidence supports the claim that it was forced auditor turnover after Andersen's demise, rather than generic state characteristics, that boosted fraud revelation.

Panel B: First stage.

- At the household-survey frequency, the first-stage coefficient on **AA Shock** is about 0.026, highly significant both in levels and in differences.

Panel C: Second stage.

- The IV coefficient on equity participation is -1.073 .

- The IV coefficient on the equity-wealth ratio is -0.600 .
- The IV coefficient on log net equity purchases is -14.533 .
- The IV coefficient on exit is $+4.905$.
- Entry and the change in the equity-wealth ratio are not precisely estimated.

Economic magnitudes from the text.

- A one-standard-deviation increase in IV-induced fraud revelation increases exit by roughly seven percentage points.
- The same shock lowers average net equity purchases by almost 8%.

Why it matters. This is the paper’s cleanest quasi-experimental evidence that the relation between fraud revelation and household stockholding is causal.

5.5 Table V: within-state households’ differences in fraud experience

Table V									
Within-State Household Differences in Fraud Experience									
This table uses the 1992 sample. The state household stock holding participation is the household lifetime experience of fraud. The main variable of interest, <i>Experienced Fraud</i> , is the average first entry experience by a household in the state of residence since the end of the household was 18. The dependent variable is <i>Equity Participation</i> in Panel A, and is <i>Equity Wealth Ratio</i> in Panel B. Prior State is the state of residence the household reported in the previous year prior to <i>Experienced State Stock Return</i> and <i>Experienced State GDP Growth</i> are the average state stock return and state GDP growth experienced by a household in the state of residence since the end of the household was 18. <i>Metrol</i> is a binary variable equal to one if the household moved to the state from another state between the previous and the current survey years. Parameter estimates are obtained by OLS and reported in column (1). OLS estimates of state and year fixed effects, when conditions are described in the text. Standard errors are presented in parentheses and clustered at the household level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.									
Panel A: Extensive Margins									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation
<i>Experienced Fraud</i>	-1.248**	-1.248**	-1.248**	-1.248**	-1.248**	-1.248**	-1.248**	-1.248**	-1.248**
<i>Metrol</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Lag(Age)</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Lag(Wealth)</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Lag(Family Income)</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>College Education</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Stock Return</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>GDP Growth</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Constant</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Observations</i>	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
<i>R²</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Panel B: Intensive Margins									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio	Equity Wealth Ratio
<i>Experienced Fraud</i>	-0.600**	-0.600**	-0.600**	-0.600**	-0.600**	-0.600**	-0.600**	-0.600**	-0.600**
<i>Metrol</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Lag(Age)</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Lag(Wealth)</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Lag(Family Income)</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>College Education</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Stock Return</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>GDP Growth</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Constant</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<i>Observations</i>	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
<i>R²</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Read: This table uses differences in lifetime exposure to fraud across households who live in the same state-year.

- In the extensive-margin regressions, **Experienced Fraud** enters negatively and significantly. In the baseline column the coefficient is about -2.138 .
- In the intensive-margin regressions, the coefficient is about -0.934 for the equity-wealth ratio.
- The results survive current-state-by-year fixed effects, prior-state controls, and variants focusing on younger households or later years.

Magnitude from the text.

- A one-standard-deviation increase in lifetime experienced fraud lowers the probability of stockholding by about 1.3 percentage points.
- It lowers the equity-wealth ratio by about 0.6 percentage points.

Why it matters. This table is powerful because it removes all contemporaneous state-level shocks. The variation comes from age and mobility histories, not from current local conditions.

5.6 Table VI: changes in equity holdings across different stock portfolios

Read: Using brokerage data, this table asks whether the effect is just a direct wealth effect on owners of fraudulent stocks.

- Even households that did *not* own fraudulent stocks cut equity holdings after local fraud revelation: coefficient about -1.065 .
- When the dependent variable excludes fraudulent stocks entirely, the effect remains: about -1.054 .
- When the dependent variable excludes stocks in the same industry as the fraudulent firms, the effect remains: about -1.154 .
- When it excludes stocks headquartered in the same states as the fraudulent firms, the effect is still negative: about -0.839 .

Interpretation. Households respond by lowering their general demand for equity, not just by dumping the fraudulent stock or other obviously contaminated firms.

Why it matters. This is the key evidence against a purely mechanical wealth effect and against a narrow informational-spillover story.

Table VI
Households' Changes in Equity Holdings across Different Stock Portfolios

This table relates the effect of fraud revelation to changes in households' equity holdings (net purchases) in different portfolios of stocks using the brokerage data from Barber and Odean (2000). The *Change in Equity Holdings* is computed as the ratio of the sum of the changes in all equity holdings during a year scaled by the value of households' equity holdings at the beginning of the year. The value of all positions is computed using share prices at the beginning of the year. *Yearly Fraud in State* is the fraud revelation intensity in the past year in a state. In column (1) we exclude households that held fraudulent stocks. In column (2) we compute the change in equity holdings excluding stocks of firms that were revealed as having committed fraud during the past year. In column (3) we compute the change in equity holdings excluding stocks of firms in the same two-digit SIC industry as the firms revealed as having committed fraud during the past year. In column (4) we compute the change in equity holdings excluding stocks of firms headquartered in the same states as the firms revealed as having committed fraud during the past year. All variables are defined in the Appendix. Parameter estimates are obtained by OLS. All regressions include a constant term and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the household level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

Dependent Var:	(1)	(2)	(3)	(4)
	Excluding Investors with Fraudulent Stocks	Excluding Fraudulent Stocks	Stocks of Firms in Industries with No Fraud Revelation	Stocks of Firms in States with No Fraud Revelation
Change in Equity Holdings				
Yearly Fraud in State	-1.065*** (0.282)	-1.054*** (0.278)	-1.154*** (0.278)	-0.839** (0.366)
Portfolio Return	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.001 (0.001)
State GDP Growth	0.174*** (0.049)	0.178*** (0.049)	0.168*** (0.049)	0.149** (0.059)
Log(Age)	-0.008*** (0.003)	-0.008*** (0.003)	-0.008*** (0.003)	-0.001 (0.003)
Married	0.006*** (0.002)	0.006*** (0.002)	0.007*** (0.002)	0.006** (0.003)
Log(Family Size)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.002 (0.002)
Year F.E.	Yes	Yes	Yes	Yes
Observations	104,932	106,256	105,143	67,407
R ²	0.005	0.010	0.010	0.006

Table VII
Fraud Revelation and Changes in Holdings of In-State and Out-of-State Stocks

This table is based on the brokerage sample. We consider only households with holdings of both in-state and out-of-state stocks. *Change in Holdings of Out-of-State Stocks* is the change in equity holdings in stocks of firms headquartered in a state other than the one in which the household resides, computed as the ratio of the sum of the changes in the holdings of out-of-state stocks during a year scaled by the value of the households' out-of-state equity holdings. The value of all positions is computed using share prices at the beginning of the year. *Change in Holdings of In-State Stocks* is computed similarly, but considering only the household's in-state stocks. The dependent variable *Difference* is the difference between *Change in Holdings of Out-of-State Stocks* and *Change in Holdings of In-State Stocks*. All remaining variables are defined in the Appendix. Parameter estimates are obtained by OLS. All regressions include a constant term and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the household level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Change in Holdings of Out-of-State Stocks	Change in Holdings of In-State Stocks	Difference	Change in Holdings of Out-of-State Stocks	Change in Holdings of In-State Stocks	Difference
Yearly Fraud in State	-1.208*** (0.431)	-0.206** (0.105)	-1.002** (0.446)	-2.088*** (0.433)	-0.518*** (0.188)	-1.852*** (0.448)
Yearly Fraud in Other States				-0.609*** (0.034)	-0.312** (0.145)	-0.559*** (0.036)
Portfolio Return	-0.001* (0.001)	0.000** (0.000)	-0.001** (0.001)	-0.000 (0.001)	0.000** (0.000)	-0.001 (0.001)
State GDP Growth	0.148** (0.074)	-0.046* (0.027)	0.194** (0.081)	-0.346*** (0.068)	-0.046* (0.027)	-0.288*** (0.073)
Log(Age)	-0.017*** (0.004)	0.003*** (0.001)	-0.020*** (0.004)	-0.016*** (0.004)	0.003*** (0.001)	-0.019*** (0.004)
Married	-0.000 (0.003)	0.007*** (0.001)	-0.007* (0.003)	0.001 (0.003)	0.007*** (0.001)	-0.006 (0.003)
Log(Family Size)	0.001 (0.003)	-0.003*** (0.001)	0.004 (0.003)	0.000 (0.003)	-0.003*** (0.001)	0.003 (0.003)
Year FE	Yes	Yes	Yes	No	No	No
Observations	53,214	53,214	53,214	53,214	53,214	53,214
R ²	0.012	0.001	0.010	0.007	0.001	0.005

5.7 Table VII: changes in holdings of in-state and out-of-state stocks

Read: This table compares how households change their in-state and out-of-state stockholdings after fraud.

- A local fraud shock reduces out-of-state holdings much more than in-state holdings.

- In columns (1)–(3), the coefficient is about -1.208 for out-of-state stocks and only -0.206 for in-state stocks; the difference is about -1.002 .
- Once the regression also controls for fraud in other states, the difference becomes even larger: roughly -1.852 in the difference specification.

Interpretation. After trust is damaged, households retreat toward familiar local assets rather than fleeing all local firms. Fraud therefore increases local bias and market segmentation.

Why it matters. This pattern is hard to reconcile with a simple story in which fraud only reveals bad information about local fundamentals. It fits a trust/familiarity mechanism much better.

5.8 Table VIII: changes in trust and fraud revelation

Table VIII
Changes in Trust and Fraud Revelation

This table relates changes in trust in various institutions in a given state between survey rounds to the intensity of fraud revelation in that state during the past year. Changes in trust are computed from the Gallup Confidence in Institutions surveys. Respondents are asked to rate their degree of confidence on a four-point scale as follows: 1 (very little), 2 (some), 3 (a lot), and 4 (a great deal). We define an individual as having high confidence in an institution if the individual reports a level of confidence greater than or equal to 3. We then compute the dependent variable as the change in the proportion of individuals that report a high level of confidence in that institution between survey rounds. The sample period is 1981 to 2009. The number of observations varies across specifications because not all questions were repeated in each survey round. We compute $\Delta(\text{Trust in Other Institutions})$ as an average of the change in trust across all institutions, except big businesses and banks, between survey rounds. All the respondent-related control variables are the change in the average logarithms of respondent age and the change in the fraction of respondents with that characteristic for all other demographic characteristics between survey rounds. All variables are defined in the Appendix. Parameter estimates are obtained by OLS. All regressions include a constant term and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the state level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

	$\Delta(\text{Trust in Big Business})$				$\Delta(\text{Trust in Banks})$
	(1)	(2)	(3)	(4)	(5)
Yearly Fraud in State ($t - 1$)	-0.722** (0.367)	-1.395** (0.542)	-1.292** (0.511)	-1.922** (0.832)	-2.207*** (0.807)
Lagged Yearly Fraud in State ($t - 2$)				1.050 (0.710)	
$\Delta(\text{Trust in Other Institutions})$			0.283*** (0.070)	0.272*** (0.068)	0.313*** (0.099)
$\Delta \text{Log}(\text{Age})$		-0.025 (0.076)	-0.040 (0.066)	-0.041 (0.070)	-0.186* (0.110)
ΔMale		0.109 (0.104)	0.126 (0.102)	0.157 (0.100)	-0.128 (0.137)
$\Delta(\text{College Education})$		-0.017 (0.051)	-0.007 (0.051)	-0.021 (0.052)	0.137 (0.108)
$\Delta(\text{High Income})$		-0.085 (0.090)	-0.118 (0.090)	-0.133 (0.084)	-0.010 (0.077)
$\Delta(\text{White})$		-0.266*** (0.089)	-0.222*** (0.079)	-0.222*** (0.077)	-0.029 (0.147)
$\Delta(\text{Black})$		0.125 (0.127)	0.091 (0.125)	0.094 (0.122)	-0.209 (0.169)
$\Delta(\text{Hispanic})$		-0.189 (0.127)	-0.168 (0.108)	-0.172 (0.107)	-0.020 (0.280)
$\Delta(\text{Republican})$		0.248*** (0.050)	0.242*** (0.052)	0.273*** (0.068)	0.159* (0.094)
$\Delta(\text{Protestant})$		0.045 (0.062)	0.022 (0.063)	0.010 (0.064)	0.102 (0.095)
$\Delta(\text{Jewish})$		0.214 (0.167)	0.179 (0.159)	0.216 (0.162)	-0.002 (0.246)
State GDP Growth (yearly)		0.189 (0.276)	0.180 (0.275)	0.294 (0.272)	1.070*** (0.388)
State Stock Return (yearly)		0.003	0.004	0.001	-0.073**

(Continued)

Table VIII—Continued

	$\Delta(\text{Trust in Big Business})$				$\Delta(\text{Trust in Banks})$
	(1)	(2)	(3)	(4)	(5)
Year FE	No	(0.025)	(0.025)	(0.024)	(0.033)
Observations	1,126	435	435	435	435
R^2	0.018	0.167	0.230	0.230	0.199

Read: This table uses Gallup data to test directly whether fraud lowers trust in financial institutions.

- For **Trust in Big Business**, the coefficient on lagged yearly fraud is between about -0.72 and -1.40 depending on controls.
- The text translates this into roughly a 7-percentage-point decline in the fraction of respondents reporting high confidence in big business for a one-standard-deviation increase in yearly fraud.
- For **Trust in Banks**, the effect is even larger: the coefficient reaches about -2.21 .

Table IX
Cross-Sectional Differences in the Effect of Fraud on Stock Market Participation

This table relies on the PSID sample. We explore cross-sectional differences in the effect of exposure to local fraud on stock market participation. *High Status* is a dummy variable that equals one if the household head is a Caucasian Protestant male. *Middle Aged* is a dummy variable that equals one if the household head is between 31 and 60. *College Education* is a dummy variable that equals one if the household head has at least 16 years of education. *High Wealth* indicates that the household's net worth is in the top tercile of the sample distribution in a year. *Low Risk Tolerance* indicates that the household head's risk tolerance score is below the sample median. As in column (3) of Table III, Panel A, the set of controls includes *Log(Age)* (except in column (2), where we include *Middle Aged*), *Married*, *Log(Family Size)*, *Log(Family Income)*, *Log(Family Wealth)*, *Lagged Equity Participation*, *State GDP Growth*, and *State Stock Return*, whose coefficients are not reported for brevity. All variables are defined in the Appendix. Parameter estimates are obtained by OLS. All regressions include the above-mentioned controls, a constant term, and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the household level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Equity Participation	Equity Participation	Equity Participation	Equity Participation	Equity Participation
Fraud in State	-0.356 (0.225)	-0.111 (0.328)	-0.167 (0.191)	-0.537** (0.228)	-0.698** (0.287)
Fraud in State * High Status	-0.513* (0.311)				
High Status	0.011* (0.007)				
Fraud in State * Middle Aged		-0.652* (0.341)			
Middle Aged		0.005 (0.006)			
Fraud in State * College Education			-1.174*** (0.370)		
College Education			0.125*** (0.008)		
Fraud in State * Low Risk Tolerance				0.115 (0.307)	
Low Risk Tolerance				0.178*** (0.007)	
Fraud in State * High Wealth					-0.093 (0.338)
High Wealth					-0.027*** (0.008)
Controls, State FE, Year FE	Yes	Yes	Yes	Yes	Yes
Observations	51,572	51,572	50,097	51,572	23,886
R ²	0.397	0.397	0.406	0.400	0.394

- The text translates this into about a 22-percentage-point decline in reported high confidence in banks.

Interpretation. Fraud revelation erodes trust in the institutions most closely connected to equity investment.

Why it matters. This is the paper's positive evidence on mechanism. The participation results are not left as a black box; the paper shows that fraud actually changes beliefs and attitudes toward financial institutions.

5.9 Table IX: cross-sectional differences in the effect of fraud

Read: This table distinguishes between trust/betrayal and standard risk-aversion mechanisms.

- The interaction of fraud with **High Status** is negative and significant.
- The interaction with **Middle Aged** is also negative and significant.
- The interaction with **College Education** is strongly negative and highly significant.
- By contrast, the interaction with **Low Risk Tolerance** is small and insignificant.
- The interaction with **High Wealth** is also insignificant.

Interpretation. The groups that react most are not the most risk averse; they are the groups the authors interpret as having higher betrayal costs. This pushes the mechanism toward trust and away from simple changes in risk tolerance.

Why it matters. The table helps the paper go beyond correlation and say something sharper about *why* fraud affects participation.

5.10 Table X: fraud revelation and changes in nonfraudulent firms' investor base

Table X
Fraud Revelation and Changes in Nonfraudulent Firms' Investor Base

This table relies on a Compustat sample of firms not revealed to be fraudulent during the sample period. The sample period is 1984 to 2009, with the exception of column (1), which relies on the brokerage sample (1991 to 1996). Data on institutional ownership are from Thomson Financial 13F. The unit of observation is firm-year. Panel A reports summary statistics. Panel B reports the estimated impact of fraud revelation on local nonfraudulent firms' investor base. In column (1), the dependent variable is the percentage change in the total number of shares of a firm held by retail investors in the brokerage sample over a year, divided by the firm's shares outstanding at the beginning of the year. *Change in the Ownership of Retail Investors*. In column (2), the dependent variable is the value of a firm's stock repurchases during a year relative to the firm's market capitalization at the beginning of the year. In column (3), the dependent variable is the percentage change in the number of institutional owners. In column (4), the dependent variable is the percentage change in the shares of a firm held by the largest institutional owner over one year, divided by the firm's shares outstanding at the beginning of the year. In column (5), the dependent variable is the percentage change in the total shares of a firm held by the top five institutional owners over one year, divided by the firm's shares outstanding at the beginning of the year. In column (6), the dependent variable is the percentage change in the shares of a firm held by institutional blockholders (institutional investors with at least 5% of the firm's shares) during one year, divided by the firm's shares outstanding at the beginning of the year. *Fraud in State* is the number of fraud cases revealed in a state during one year, divided by the number of listed companies headquartered in the state. We control for firm market capitalization at the beginning of the year (*Firm MktCap*) and firm past return (*Firm Past Return*). All variables are defined in the Appendix. Parameter estimates are obtained by OLS. All regressions include a constant term and fixed effects as indicated in the table, whose coefficients we do not report. Standard errors are presented in parentheses and clustered at the firm level and corrected for heteroskedasticity. ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

Panel A: Summary Statistics			
	<i>N</i>	Mean	<i>SD</i>
Change in the ownership of retail investors	14,439	-0.0001	0.0006
Repurchases	91,485	0.012	0.058
Change in the number of institutional owners	78,363	0.159	0.769
Change in the ownership of the top institutional owner	78,130	0.009	0.046
Change in the ownership of the top 5 institutional owners	78,130	0.025	0.091
Change in the ownership of institutional blockholders	45,665	0.023	0.112
Firm MktCap	91,485	5.237	1.980
Firm Past Return	91,485	0.155	0.777
Fraud in State	91,485	0.003	0.004

(Continued)

Table X—Continued

Panel B: Changes in Nonfraudulent Firms' Ownership

	(1)	(2)	(3)	(4)	(5)	(6)
	Change in the Ownership of Retail Investors	Repurchases	Change in the Number of Institutional Owners	Change in the Ownership of the Top Institutional Owner	Change in the Ownership of the Top Five Institutional Owners	Change in the Ownership of Institutional Blockholders
Fraud in State	-0.002* (0.001)	0.058* (0.033)	0.573 (0.501)	0.088* (0.050)	0.219** (0.093)	0.318** (0.147)
Firm MktCap	0.000*** (0.000)	0.002*** (0.000)	-0.039*** (0.001)	0.0004*** (0.0001)	0.002*** (0.0002)	0.001** (0.0002)
Firm Past Return	0.000 (0.000)	-0.001*** (0.000)	0.031*** (0.010)	0.002*** (0.0002)	0.006*** (0.001)	0.009*** (0.001)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,439	91,485	78,363	78,130	78,130	45,665
<i>R</i> ²	0.010	0.014	0.023	0.007	0.017	0.015

Read: This table studies spillovers onto innocent firms headquartered in states hit by fraud.

- Retail ownership of local nonfraudulent firms falls: coefficient about -0.002 .
- These firms increase stock repurchases: coefficient about $+0.058$.
- The number of institutional owners does not significantly rise.
- But ownership becomes more concentrated among large institutions:
 - top institutional owner: $+0.088$,
 - top five institutional owners: $+0.219$,
 - institutional blockholders: $+0.318$.

Interpretation. Retail investors withdraw, firms buy back stock, and the remaining ownership becomes more concentrated among large institutions. This is consistent with a higher cost of capital for nonfraudulent local firms.

Why it matters. The paper's externality argument is not purely conceptual. Table X shows firm-side consequences in the composition of the investor base.

6 Short summary

- **Question.** Does corporate fraud reduce household demand for equity by lowering trust?
- **Design.** Match household location to local fraud revelation using fraud-enforcement data, PSID household surveys, brokerage-account records, and Gallup trust surveys.

- **Main reduced-form result.** Local fraud revelation lowers stock market participation, raises exit, lowers entry, reduces net equity purchases and the equity share of wealth, and shifts portfolios toward fixed income.
- **Main identification result.** Exogenous fraud revelation induced by Arthur Andersen’s collapse causes similar or larger reductions in participation.
- **Mechanism result.** Fraud reduces trust in big business and banks; the effect is strongest for groups associated with higher betrayal costs, not for groups that are simply more risk averse.
- **Spillover result.** Households cut holdings even when they did not own fraudulent stocks and even in nonfraudulent firms. Nonfraudulent local firms lose retail investors and end up with more concentrated institutional ownership.
- **Bottom line.** Corporate fraud creates a negative trust externality that depresses equity demand beyond the fraudulent firm itself.

7 Conclusion

- **Core conclusion.** Corporate scandals decrease household stock market participation, and the evidence strongly suggests that this happens because fraud undermines trust in financial markets.
- **Economic intuition.** Buying equity requires not only tolerance for risk but also confidence that firms, managers, auditors, and intermediaries are not cheating. When fraud is revealed, that confidence falls, so households step back from the market.
- **Why the paper matters.** The paper broadens the cost of corporate fraud from a firm-level event to a market-wide externality. Fraud by one firm can reduce demand for equity more generally and thereby raise financing costs for innocent firms.
- **What to remember.** The paper is one of the clearest demonstrations that trust is an economically important state variable in household finance.